

Dave Dellenbaugh's **SPEED & Smarts**™

The digital newsletter of how-to tips for racing sailors **ISSUE 150**



150 fast & smart tips!

Welcome to the 150th edition of *Speed & Smarts* and our special, expanded 25th anniversary issue! The newsletters look a little different now than they did in the past, but we still try to pack each one full of helpful tips about strategy, tactics, speed, racing rules, boathandling and more!

This newsletter contains 150 tips to help improve your performance. They have been selected from the previous 149 newsletters and cover a wide range of racing topics (*see list at right*). The tips are brief, but each one could be the subject of an entire issue, and hopefully they will give you a good bit of 'food for thought' for the near future.

The first tip is, simply, 'Have fun,' while the final tip suggests that you try to 'Learn something from every race.' Though these two ideas may not help you win a race today or tomorrow, both are key for long-term success in the sport. And they describe my personal philosophy about what's important in sailing. I hope you enjoy this issue!



Have fun!

If you want to improve your race results, be patient and willing to learn. Improvement takes time and hard work, and this won't be easy unless you're having fun. As Dr. Stuart Walker once said, "Winning may be the object of the game, but it is not the object of *playing* the game." The reason we play the game is for challenge and fun. So while you're competing, remember what you enjoy about our sport.

SPECIAL ISSUE!

TOP 150 TIPS

Introduction	1
Before the Race	2
Starting	4
Strategy	7
Windshifts	8
Puffs & Lulls	10
Boatspeed	11
Light Air	14
Downwind	15
Boathandling	17
At Marks	19
Tactics	21
Learning	23

Play the odds.

Sailboat racing is all about probability. We never know exactly what's going to happen with the wind and other variables. That makes the sport fun and challenging, but it also means we must play the odds by choosing strategies and tactics with the best chances of success. Whether you are maneuvering on the starting line, picking a side to play on the first beat, or deciding whether to cut inside a few boats at the leeward mark, your goal should be to consistently choose actions with a high probability of success. You don't need perfection – you just want to avoid most actions that fall into the high-risk category (*right*).



Before the Race

Have 'No Excuse to Lose.'

The title of Dennis Conner's popular book offers great advice for your mental and physical preparation. The basic idea is this: Before the start of every regatta or race you should eliminate any reason why you might not win that race.

Paul Elvström, the Great Dane, offered similar advice when he said, "The good sailors all know how to race very well, but the champions have won the regatta before the racing begins." In other words, he and Conner both agree that preparation (of boat, crew, mind, etc.) is absolutely key to racing success.

Be a detective.

To be successful, sailors have to be like detectives. In other words, they must be skilled at finding clues to solve a mystery (of what will happen next on the race course). We never know for sure how the wind will shift or where the best pressure will be, but many clues around the race course can help us with these predictions. And, of course, the sailors who are best at predicting changes in the wind are usually at the front of the fleet. So keep looking for clues!



Resolve to avoid contact. Whether you have the right of way or not, hitting (or touching) another boat is almost never in your interest because it brings a number of risks: 1) that you will be penalized under rule 14 (Avoiding Contact); 2) that you will be disqualified for an incident in which you wouldn't have been wrong if there was no contact; and 3) that you will cause damage or, worse yet, injury. My suggestion is that before a race or series you make a conscious decision as a team to avoid all contact (if possible).

Stand up for a better view.

When you're trying to see what is happening up the course, the higher you can get the better. Gaining height allows you to see farther and gives you a more accurate picture of the water surface to windward. This is especially important when the wind is light and patchy and you are looking for subtle clues. The easiest way to get a better view of the wind is simply to stand up on a high part of your deck or cabintop. This is easy to do before (or during) a race, and the extra few feet of height it gives you will provide a much better view of the wind.



Set goals.

In sailing, your goal might be to win a big regatta or simply to improve your performance in the next race. In either case, the best way to get there is usually by starting with small, sure steps. Begin by setting manageable, achievable performance goals rather than pie-in-the-sky, results-oriented goals. For example, it's easier to work on your tacks than to win the next regatta. Set goals that are within your control, not ones that are dependent on your competitors.

Track the wind shift history. The best clue about what the wind will do in the near future is usually what it has done in the recent past. If the wind direction has been oscillating back and forth about 20° during the hour before the start, for example, there's a good chance this will keep happening during the race. That's why it's important to study the history of wind shifts before the start (and during the race). Do this long enough to discern a pattern and then include this info in your starting and first-beat strategy.

Check the wind from starboard tack. When you go head to wind before the start, do it on starboard tack. That way you will maintain the right of way over all other boats in the starting area, and you will have a consistent process for taking wind readings.

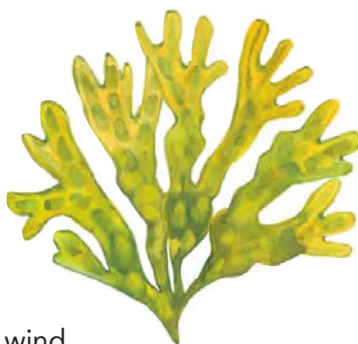


JH Peterson photo

Before the Race

Get rid of weeds.

It's definitely slow to sail around the course with weeds on your rudder, keel or centerboard. Even when you just *think* you might have weeds, this often affects your boat-speed. So don't leave anything to chance. When you get close to your start, go head to wind and back your mainsail until you are going in reverse. Do this near enough to the start so you minimize your chances of catching any more weeds, but far enough ahead of time that it won't interrupt your starting strategy (I usually do a 'back-down' with about 4 or 5 minutes to go).



Warm up your team. You wouldn't begin playing a soccer game without stretching your muscles and kicking the ball at least a few times. So don't start a sailboat race cold either. Get out to the course area early enough so you will have time to get ready. Start by doing some easy tacks and jibes. Once everyone starts to get warmed up, take a break and do some stretching in the cockpit or on deck. Pay particular attention to the muscles you normally use while sailing. Then do a few more rigorous tacks and jibes, plus other maneuvers like stopping and starting, 360-degree turns, spinnaker sets and so on. If possible, practice every maneuver you may need in the race, slowly building up to race-course intensity.

Get rid of unnecessary weight.

Heavy is slow, and the last thing you want to do in most boats and conditions is carry extra, non-essential weight around the race course. To prevent this, go through your hull, spars and equipment and get rid of all unnecessary fittings. When you push off from the dock, leave behind extraneous clothes, gear, tools and so on. Once you're out racing, the most common extra weight is water in your bilge. Every gallon weighs about eight pounds, and what looks like just a little water in your boat can easily be several gallons. So keep your boat dry (but be sure you have enough drinking water)!



Find the windward mark. The location of the first windward mark can be a critical factor in your plan for the start and first beat. If the mark is well to the right of the wind, for example, this favors playing the right side of the first beat and moving closer to the race committee boat end at the start. Therefore, make sure you know where they put the top mark before you commit to a starting plan.

Read the sailing instructions.

The sailing instructions (SIs) are your guide to what will happen on the race course, so make sure you know these inside and out. Read them before the first day of racing and review them with your team at your pre-race crew meeting. I like to present the SIs to my team as the 'Top 10 things you should know about the SIs.'

Each morning be sure to check the official notice board before you leave the dock for any changes to the SIs. Also, keep a copy of the SIs (plus a current rulebook) handy while you're racing, just in case.

Tune up with another boat.

It's hard to over-state the value of sailing upwind near another boat and working on speed. This is a great way to fine-tune your steering, gear shifting and communication for each day's conditions. So find a tuning partner and get fast!



Hold a team meeting.

Getting your crew together before the race can be extremely valuable for preparation and for building team spirit. This should be done well before the first warning signal, preferably on the way out to the course. Its purpose should be to share information that crewmembers need to sail the race. Possible topics include:

- Crew roles (assignments, communication)
- Race procedures (e.g. sailing instructions)
- Weather information (forecast, strategy, etc.)

The goal of this meeting is to get all crewmembers involved in the upcoming race and thinking about their roles. The bigger your crew, the more important it is to have an organized meeting. Be sure to hold this gathering before the starting frenzy begins!

Starting

Congestion = Risk!

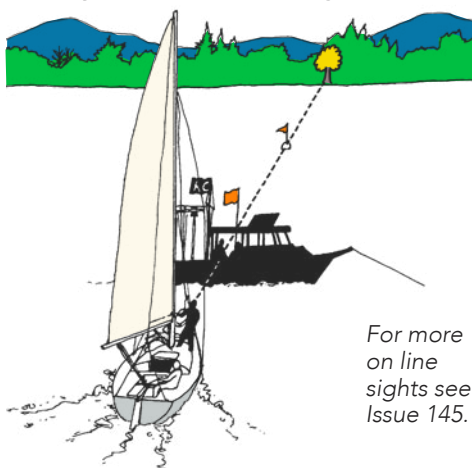
If you want to make starting really difficult, put yourself in a densely populated part of the line. As a rule of thumb – the bigger the crowd, the smaller your chances of getting a good start. So stay away from other boats as much as possible.

Practice time & distance.

When the starting gun goes off, your goal is to be close to the line at full speed. This is not as easy as it sounds, for several reasons: a) the wind is always changing; b) the starting line can be hard to find; and c) it's tough to predict exactly how your boat will accelerate. For these reasons, it's very helpful to practice a bunch of timed runs. Try a couple of these at the starting line before each race or, on a practice day, pick a buoy, set your timer and try to be there at full speed at zero.

Use a line sight!

One extremely valuable tool for starting is a 'line sight' (or 'range') using the pin end of the line and a point on shore behind it. Get a line site for every start (if possible) by sailing just outside the committee boat and looking through the pin end toward the land beyond. By lining up these two points as you approach the start, you'll know exactly where the starting line is.



For more
on line
sights see
Issue 145.

Go for a high-percentage start.

The start of a race is not usually the time to take a lot of risks. With the whole race in front of you, your goal should be to get a good, clear-air start that will allow you to follow your strategy. Because of this, you should generally stay away from the ends of the line. The areas near the pin and committee boat are usually congested and offer a low probability of accomplishing the goals above. Unless you are really good at starting tactics and boathandling, it's probably better to start where there aren't so many boats. Being closer to the 'unfavored' end with clear air is often better than being in the third row at the favored end.



Don't get too far from the line. The farther you are from the starting line, the harder it is to see the line and what's happening on the race course to windward, and the more likely it is you will be late. This applies both before and during your starting sequence. So when you are waiting or maneuvering for the start, stay generally in the middle of the line (not way off to one side). And when you make your final approach to the start on starboard tack, keep your bow poked ahead of the pack.

Go with the flow. There is typically a counter-clockwise movement of boats around the starting area while competitors jockey for position. Starboard tackers run down the line toward the pin with port tackers to leeward sailing back toward the RC boat. It's usually best not to buck this trend (or you could have rule problems). But also be careful not to get pinned or blocked by nearby boats going in the same direction.

Keep watching the wind. Most sailors do a good job of getting out to their starting area early and carefully tracking windshift patterns. But once the race committee fires the warning gun, those same sailors often get caught up in starting tactics and stop thinking about the wind. This is problematic for a couple reasons: First, the wind doesn't stop shifting during your starting sequence. And second, the wind shifts that occur in the last few minutes before your start are the most critical ones for knowing what you should do right after the gun. So while you are sailing around the starting line, keep taking wind readings by going head to wind or sailing closehauled on starboard tack. Make sure you do this in clear air and avoid any risk of fouling other boats.

Starting

Keep a good lookout.

The starting area is the most crowded part of any race course, so make sure to watch where you are going. Appoint someone on your boat to keep an eye out for other boats at all times before your start.

Take a proper penalty.

One big risk of not keeping a good lookout is fouling another boat. If you do break a rule, note that you can (and must) take your penalty as soon as possible by getting well clear of other boats right after the incident and then promptly making two turns. Because you can take your penalty before the start, pre-start fouls are usually less costly – you can often take your penalty and get back into the starting line-up before the gun.

Keep an eye on the RC.

The race committee's job is to run the race and communicate details about this with the sailors. Don't turn your back on this helpful information. Take careful note of actions such as the following:

- the precise timing of the signals;
- the course axis and distance to the first mark;
- the preparatory signal (i.e. does it signal a starting penalty?); and
- whether the RC flies an X flag to indicate that a boat(s) was OCS.

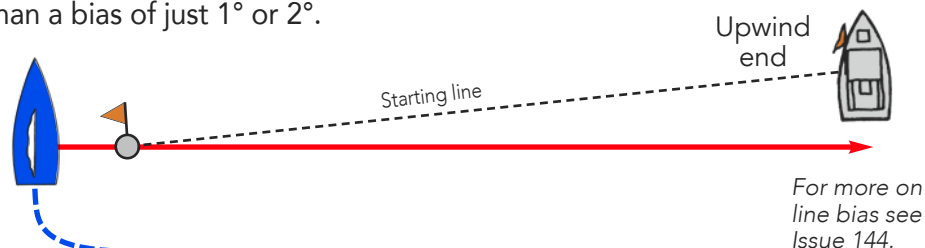
Missing information such as this could make the difference between winning or losing a race.

"You don't have to 'win' the start in order to have a successful race."



Determine the starting line 'bias.'

When you're figuring out where to start, one big factor is the angle of the starting line relative to the wind direction. If the line is perpendicular to the wind we call it 'square' and all boats starting on the line are even. If the line is not square it is 'biased,' and one end is farther to windward. By starting at that end you can get a free head start, so it's key to figure out which end is more upwind – and by how much. Of course, a bias of 10° to the RC boat end is much more significant than a bias of just 1° or 2°.



Don't confuse 'upwind' with 'favored.' When somebody asks you which end of a starting line is 'favored,' they usually want to know which end is more upwind. But 'upwind' and 'favored' don't always go together. If all other strategic factors are equal, it would be better to start near the upwind end. However, racing conditions are seldom equal or steady across the course, so finding the favored end usually involves more than just determining which end is farther upwind.

Before deciding where to start, you must also consider current, wind direction and pressure across the course, plus the position of the first mark. Often these factors favor the end that's not farther upwind. For example, if the pin end is to windward but there's better pressure on the right side, the pin end may not be favored. For this reason, be sure to distinguish between an end that is 'favored' versus 'more upwind.'

Start with a 'cupcake' to leeward.

When you come off the line at the start, what's your biggest worry? For many sailors, it's the proximity and speed of the boat on their leeward side. If that boat is too close, or if she is fast or points high, it will be tough to maintain clear air.

Sometimes it's actually possible to position yourself just to windward of a 'cupcake' (a slow boat). When making your final port-tack approach before the start, look for a slow boat that's already on starboard tack near where you want to start. Tack right behind (and just to windward of) them, and hold your position there until the gun.



Move to the right in your space. The most important factor in getting a fast start with clear air is having enough space on your leeward side so you can put your bow down and rumble after the gun. That's why as you approach the line you should try to move from left to right (i.e. to windward) in the space between the boats on either side of you. By moving farther away from the boat to leeward you will have more space to accelerate and less chance of falling into their bad air. And by getting closer to the windward boat, you'll have more control over them and hopefully be able to pinch them off after the start.

Starting

Don't be greedy! As you slowly approach the start line on starboard tack, try to create some space between you and the boat to leeward (so you won't get pinched off after the start). But don't go for more space than you need. If you create too big of a hole to leeward, you'll attract boats that are still looking for a front-row spot on the line. To avoid this, carve a hole to leeward that is just big enough to get a good start, but just small enough so it won't attract competitors.

Don't start by the gun. Don't use the starting sound signal as a measure of when to trim in and go:

1) Even if the starting gun is loud and exactly on time, it could take a second or two to travel down the line to you;

2) The visual starting signals are official. The sound signal is unofficial so it may be late or it may not be made at all; and

3) In order to be up to full speed at the start you must 'trim in and go' before your watch gets to zero.



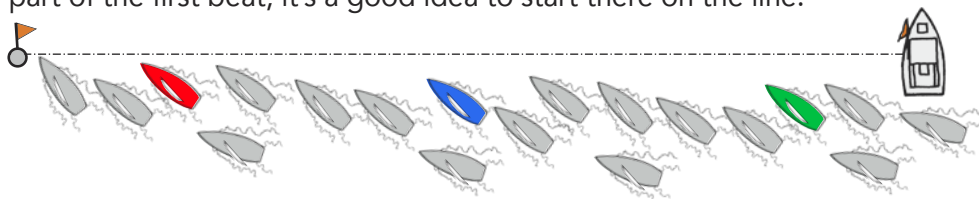
Set up in the front row. When making your final approach on starboard tack, it's normally better to be in the front row early:

1) It's usually easier to begin in the front row and defend the space around you than it is to come from behind later and try to squeeze into the front row; and

2) In the front row you have a much better view of the line, your line sights and wind up the course.

Start where you want to go.

As a rule of thumb, a boat's starting position usually corresponds with where she ends up on the first beat. Boats that start near the pin end of the starting line, for example, usually play the left side of the beat. Boats that start near the committee boat usually end up on the right side of the beat. And boats that start in the middle are usually somewhere between the left and the right. Unless there is a lot of line bias or a large wind shift after the start, boats are seldom able to start at one end of the line and play the opposite side of the course. Therefore, when you favor a certain part of the first beat, it's a good idea to start there on the line.



Luff on a closehauled course.

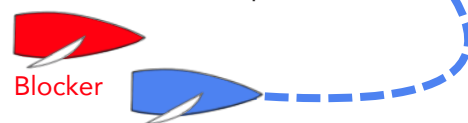
When you luff your sails to slow as you make your final approach on starboard tack, don't try to sail head to wind (except when you still have a lot of momentum). If you get too slow and too high, it will be hard to accelerate for the start:

1) you'll have to turn down to a closehauled course, which is difficult when going slowly; and

2) you can't trim in your sails until you bear off. So keep the boat on (or near) a closehauled course while luffing so you can trim and accelerate without turning the rudder.

Don't get pinned on port.

When you're making a port-tack approach to the start, be sure you are able to tack onto starboard at the place and time you want. The biggest danger is getting blocked by a port tacker to windward or astern who makes it difficult or impossible to tack when you want. If you get into this position, take proactive steps to escape the blocker as soon as possible.



Maintain steerageway. One reason why you shouldn't sail close to head to wind (for very long) is that you'll slow down and eventually lose steerageway. When that happens, it can take a long time to get the boat going again, especially with bigger, heavier boats that don't accelerate very well. For these boats a critical rule of thumb is to maintain steerageway as you approach the starting line. Practice this by sailing as high and slow as you can while still keeping just enough speed to turn the boat. This will help you manage your time and distance to the line.

Work hard for 3 minutes. The first few minutes after your start are arguably the most critical moments of any race. That's when some boats punch out ahead of the fleet into clear air and others get sucked back into the pack. During this time it's important to focus and work hard on speed. For example, this is when your entire team should be hiking absolutely as hard as they can.

Bail out early. One of the biggest mistakes sailors make when they get a bad start is staying on starboard tack too long in bad air. The longer you wait to clear your air, the farther you will get behind the boats that are sailing in clear air. Minimize damage by tacking as soon as you have an opening, even if this means going the 'wrong way' for a short time.

Strategy

Strategy first!

No matter where you are on the race course, strategy almost always comes before tactics. First you make a strategic plan for where you want to go (in the absence of other boats), and then you use tactical moves (if necessary) to help you implement your strategy in the midst of a fleet.

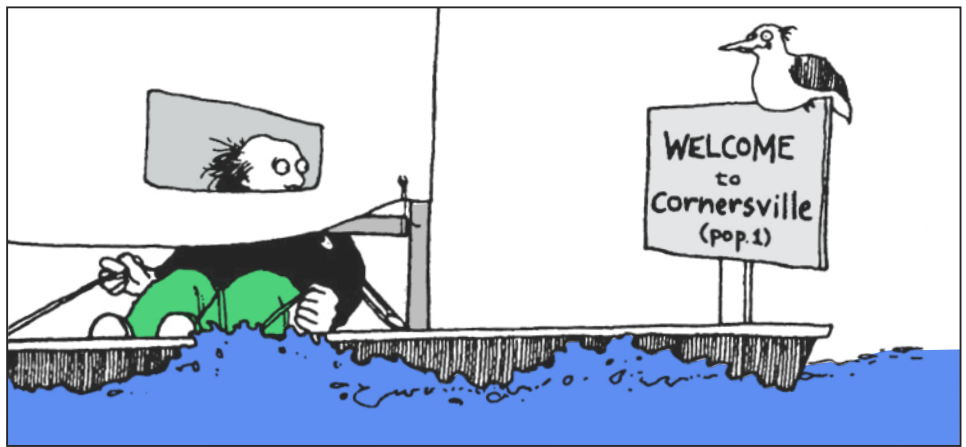
Strategy
↓
Tactics

Assess your risk tolerance.

When you race sailboats, you have to make a lot of decisions. And each decision carries a certain amount of risk. For example, on a beat you have to decide whether to play the right side, the left side, or the middle. And if you go to one side, how far should you go? How much risk will you take? To help make these decisions, have a discussion with your team before the race or series about your tolerance for risk. Then use this discussion to guide your race-course choices.

Go up-current first.

When the current is not aligned with the wind, look for strategic opportunities. If the wind and current are steady, it won't matter which tack you sail first. But when it's likely that the conditions will change, a high-probability move is sailing the upcurrent tack first. The tack that takes you more into the current is also usually the longer tack. Sailing this upcurrent tack won't by itself get you to the mark faster, but it will leave you in better position to handle future shifts.



Avoid corners and laylines.

On beats and runs, the corners and laylines are usually bad places to be because: 1) you may overstand the mark and sail extra distance; 2) you will lose to other boats if you get a lift or a header; and 3) other boats may tack or jibe on you and give you bad air all the way to the mark (or make you do two extra tacks or jibes and overstand the mark). This is why it's usually better to stay away from laylines, especially when you are relatively far from the windward or leeward mark.

Make a plan for the next leg.

Before you round a mark, put together a strategic plan for the ensuing leg. This is your plan for how to use wind shifts, puffs and current to sail that leg as quickly as possible. Your strategy can be as simple as left or right – you just need some guidance for how to begin sailing that leg.

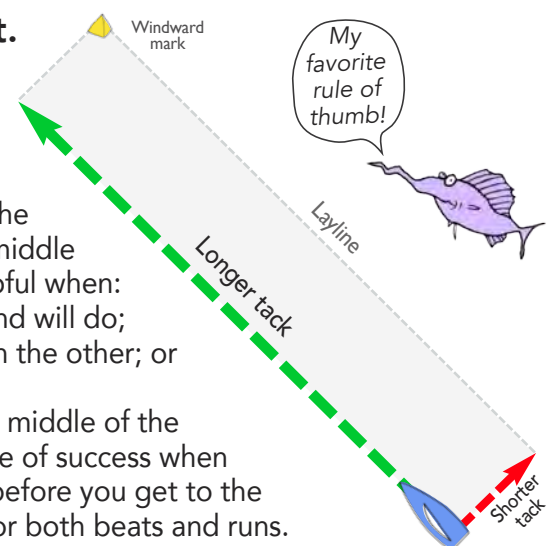


Sail the longer tack first.

One way to avoid getting to a layline too early is by using this rule of thumb: Sail the tack on which your bow is pointing closer to the next mark. This is the tack that takes you toward the middle of the course. It's especially helpful when:

- 1) You're not sure what the wind will do;
- 2) One tack is a lot longer than the other; or
- 3) You are far from the mark.

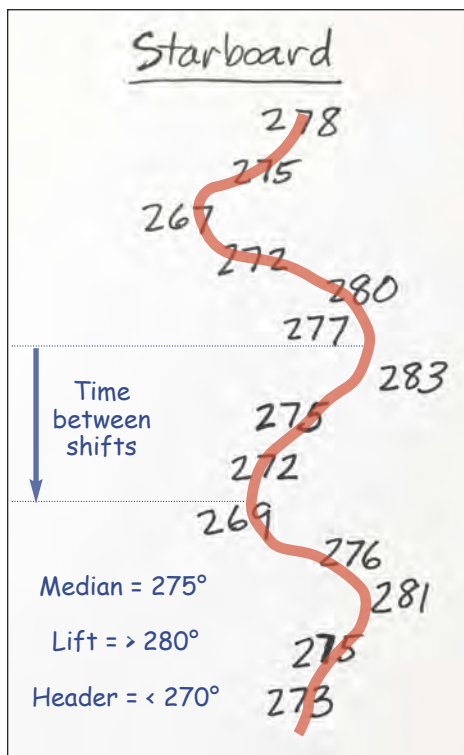
Sailing toward the mark and the middle of the course gives you the best chance of success when conditions are likely to change before you get to the mark. This strategy is valuable for both beats and runs.



Windshifts

Gather good wind data.

When you collect wind data before your start, be sure to keep good track of the range of your headings on starboard tack. That's because when you come off the line after the gun, you are almost always on starboard. The first shift after the start is the most important, so you need to know right away if you are headed or lifted. The only way to determine this is by knowing if your starboard-tack compass heading is relatively high or low.



Pick a median.

When the wind direction is oscillating, it's important to figure out your median (average) heading on each tack. This is usually the heading that is half way between the biggest header and the biggest lift you've seen on that tack. It's key to know this number because your strategic choices hinge on whether the wind is to the left or right of this median. Remember the average wind direction can change during the race.

Sail toward the next shift. One thing we know about the wind is that it's almost always shifting. If you can predict the direction of the next shift, a key strategic rule of thumb is to sail toward that shift. So if you think the wind will shift to the right you should sail on port tack; if you expect it to shift left, sail on starboard.

In an oscillating breeze, sailing toward the next shift means tacking on the headers. In a persistently shifting breeze, sailing toward the next shift means digging in to the favored side. In both cases, this strategy works for boats racing upwind because when the wind shifts they end up on a higher 'ladder rung,' ahead of boats that didn't sail toward the shift.



Oscillating or persistent? On most days, the wind is either shifting back and forth (oscillating) or shifting slowly in one direction (persistent). At each moment of a race, you must decide which way you are going to play the wind: oscillating or persistent? This choice influences many of your strategic and tactical moves and could greatly affect your success. For example, if you think the wind is shifting persistently, you should keep sailing into a header. But if it turns out the wind was oscillating instead, you should have tacked on that header.

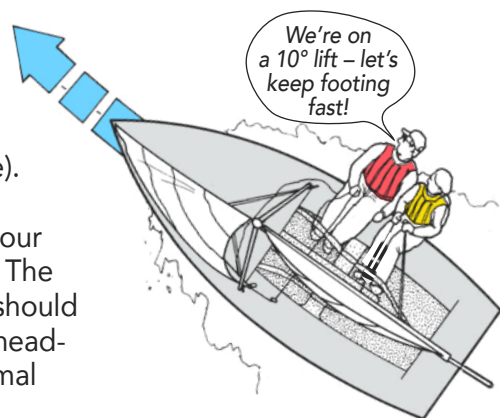
Tack on headers. In an oscillating breeze, the most basic rule of thumb is to tack when you get headed. By doing this you will stay on the lifted tack and sail the shortest distance to the windward mark. Don't tack on a header if it is caused by a lull, if you need to sail farther into the shift to get more pressure, or if it's the last shift before the mark (see right).

Play the last shift as persistent.

If you are tacking on the headers in an oscillating breeze, and you get close enough to the windward mark that the wind will not shift again on that leg, don't tack on the header as usual. Instead, treat the last oscillation as if it is a persistent shift (because it won't shift back the other way again on this beat). Keep sailing into the header (below your median heading) until you are able to tack and fetch the mark.

Sail fast on lifts.

In a shifty wind, it's usually better to sail slightly faster than normal while you are on the lifted tack (which is hopefully most of the time). This will bring you to the next shift (header) sooner, and it maximizes your VMG in the median wind direction. The more you are lifted, the faster you should sail. When you are at your median heading, you should be sailing your normal (optimal) upwind speed and angle.



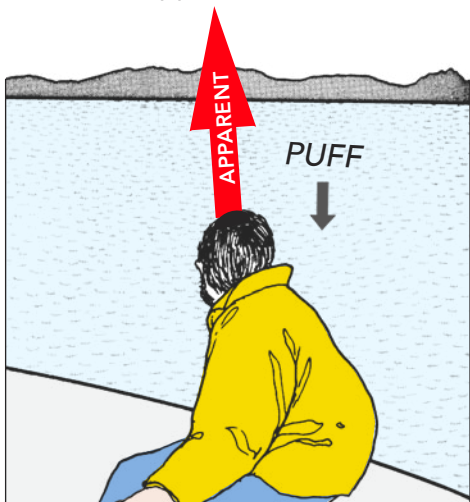
Windshifts

Beware of 'pressure shifts.'

Most of the windshifts you experience on the race course are just that: changes in the direction the wind is blowing. Sometimes, however, what appears to be a shift in wind direction is actually caused by a change in wind speed.

For example, when you sail into a lull it temporarily seems like a header, and a puff at first appears to be a lift. But don't treat these pressure shifts like 'real' ones. If you tack on a pressure header, for example, you'll be tacking in a lull, which is usually not a good move.

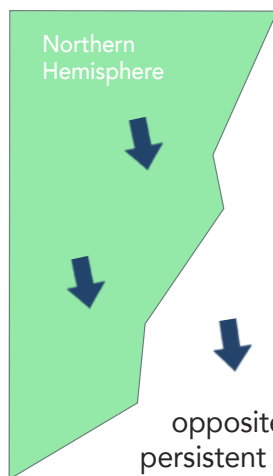
When you're not sure whether you just got a pressure shift or a 'real' one, wait a few seconds and see what happens.



Where to look for puffs.

Sailors trying to find their next puff will often look straight upwind (in the direction of the wind ripples), but this is too far aft. The puffs that are directly to windward of you at this moment will reach you only if you stop right where you are. If you keep moving forward, those puffs will pass behind you.

The place to look for the puffs that will hit you is in the direction of your *apparent wind* (i.e. the wind you feel on your body). This direction is where your masthead fly and shroud telltales point.

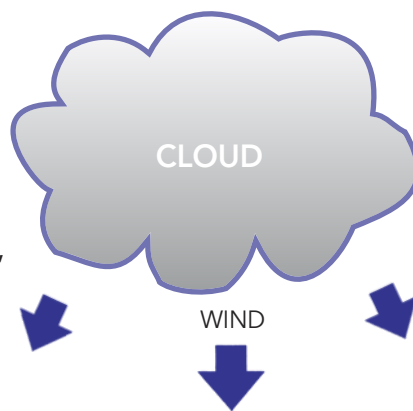


Look for shifts near land.

When the wind is blowing along (or at an angle to) the shore, you will likely see a change in direction near shore because friction over land slows and bends the wind. This geographic effect is more pronounced as you get closer to shore. If you're in the Northern Hemisphere and sailing toward a shore on starboard tack you will get headed as you near it. If you're on port tack you will get lifted. The opposite is true in the Southern Hemisphere. Treat this like a persistent shift; i.e. sail into it but be careful not to overstand.

Watch clouds for shifts.

Clouds can have a significant effect on wind pressure and direction. The bigger the cloud, the more potential there is for changes in the wind. A dark, rainy cloud (right) is likely to have stronger breeze fanning out ahead of its downwind edges, so it's usually good to sail upwind toward this type of cloud. Puffy white clouds tend to have less wind speed underneath and don't affect wind direction as much.



Look for wind shear.

Wind shear (a change in wind direction as you go aloft off the water) is often an indicator of a windshift to come. If you notice the wind at the top of your mast is sheared to the right, for example, it's a good bet that you will see a similar right shift at surface level sooner or later. Consider this possibility in your strategy for the next leg(s).

Tack when headed to the median.

When the wind is oscillating, most sailors know they should be tacking on headers. But often boats sail too far into the wind shift before tacking. So when, exactly, is the right time to tack?

The goal in an oscillating breeze is to sail on lifts all the way to the windward mark. To do this you should tack roughly when you are headed to your median course on each tack. If you wait until you are sailing below the median, you'll spend that time on a header (not a lift).

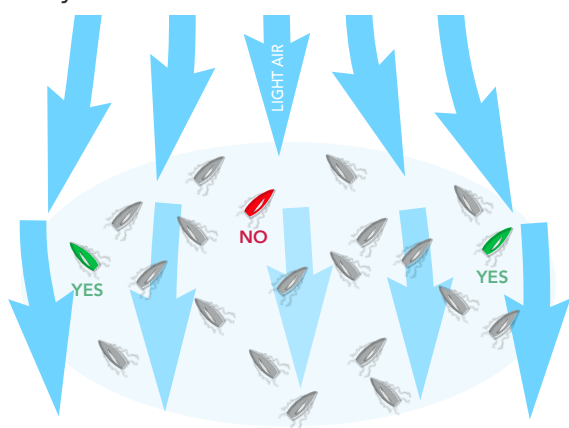


Puffs & Lulls

In light air, avoid the middle.

When racing in a large fleet, have you ever sailed up the middle of the first beat and been passed by boats on both sides? This is one time when staying near the middle may not be so conservative.

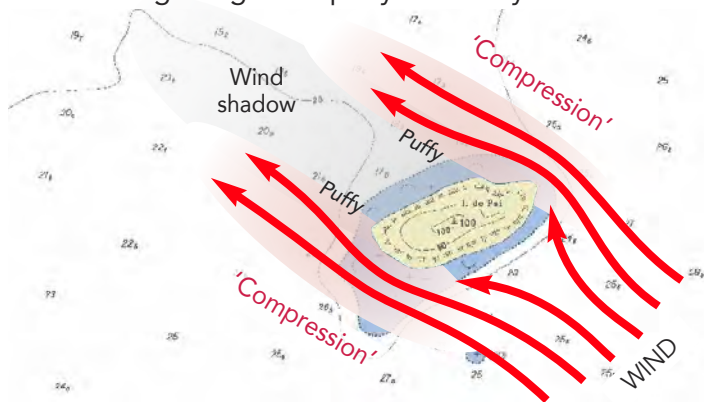
In lighter wind, there is often less air in the middle of the course than on either side. A large fleet acts like a snow fence, and the wind has a hard time getting through all the sails. As a result, the breeze tends to bend around and lift up over the bulk of the fleet. Even if you go the 'wrong' way, the extra pressure you find on the edge will often more than make up for any distance you lose due to windshifts.



Set up for lulls, not puffs. On a day when the wind is puffy, it's hard to know how to set up your sails and tuning. If your sails are too full, you'll be slow in the puffs, but if they're too flat you will struggle in the lulls. The best choice is usually to err on the side of tuning for lulls. That's because it's relatively easy to go fast in the puffs – being in a puff is great, and you can live with being overpowered temporarily. But in the lulls you need lots of power, and if you don't have it you will be noticeably slower than your competition.

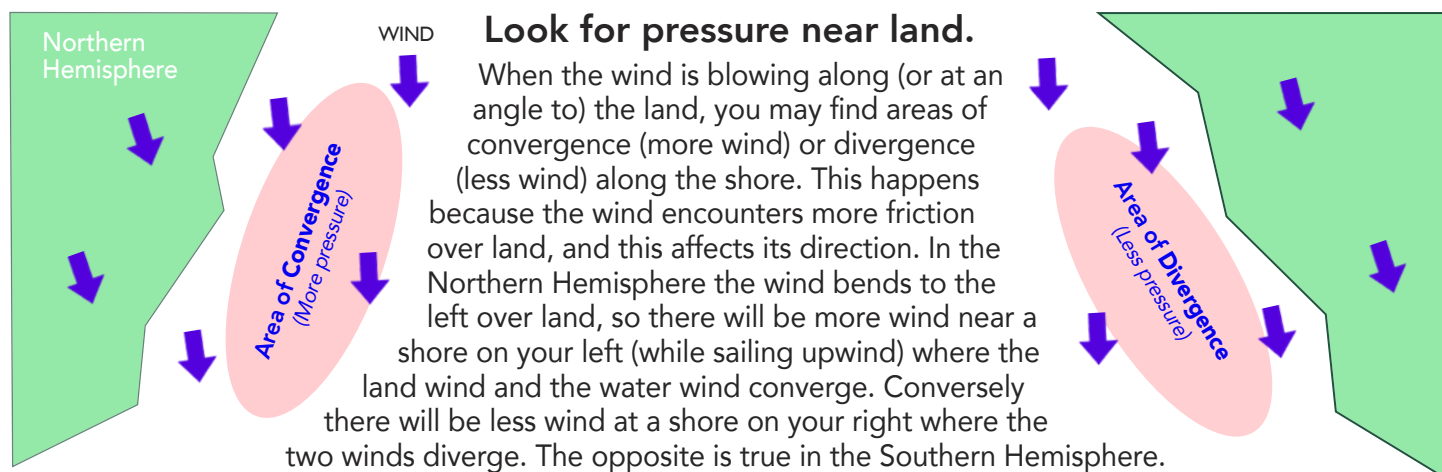
Watch for pressure changes near bluffs.

When you're racing near high points of land, watch out for two effects on wind speed. On the leeward side of the bluff, of course, is a wind shadow that you should avoid. But along the side of the bluff you will often find more pressure as the wind flow compresses to get around it. This is usually a good place to be, but watch out for getting too close to the wind shadow. You'll know you are in the transition area when the wind starts getting more puffy and shifty.



Identify the shift pattern. When you look across a race course, it's difficult to see changes in the wind direction, but relatively easy to see variations in wind pressure. Often you can use wind pressure to help identify the wind shift pattern. For example, if you see puffs and lulls up the beat, the wind shift pattern is very likely to be oscillating.

Focus on down-shifting. It's important to 'shift gears' (i.e. change your sail/boat trim settings) when the wind speed changes, and this is certainly a priority for sailing in puffs and lulls. Most sailors do a relatively good job of shifting up when they get a puff (or at least the extra wind speed makes them look good). But they aren't so good about shifting down when the pressure drops. This ability often separates the best sailors from the rest, so sailors should focus on this.



Look for pressure near land.

When the wind is blowing along (or at an angle to) the land, you may find areas of convergence (more wind) or divergence (less wind) along the shore. This happens because the wind encounters more friction over land, and this affects its direction. In the Northern Hemisphere the wind bends to the left over land, so there will be more wind near a shore on your left (while sailing upwind) where the land wind and the water wind converge. Conversely there will be less wind at a shore on your right where the two winds diverge. The opposite is true in the Southern Hemisphere.

Boatspeed

Point higher by going faster.

The ability to point high is great for tactics, strategy and speed, and that's why every sailor works hard to have better height when racing upwind. But you can't just aim your boat closer to the wind and expect the boat to go that way. Pointing ability is inextricably connected to speed, so if you want to point higher you have to start by going faster. This gets the water flowing more effectively over your foils and generates better lift.

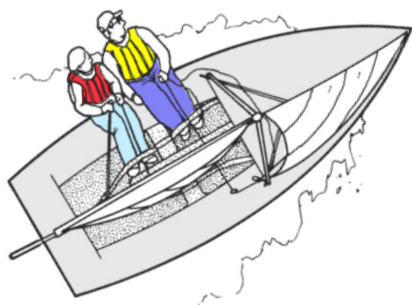
Keep trying to trim harder.

One key to going fast is the ability to feel what your boat and sails are telling you. When I'm driving and trimming the main, I follow this rule of thumb: If the boat feels good (i.e. it feels lively, like it's going fast through the water), I trim the main a little harder and point the bow up slightly higher. If the boat still feels good, I do this a little more.

When I feel the boat start to slow, I ease the mainsheet a touch and bear off a bit. Once I'm going fast, I start trimming in again. This is a continuous cycle of using the feel of the boat to keep it going fast and always trying to point high.

Steer by the middle telltales.

When you steer by the jib telltales, use the ones that are half way up the luff. These will give you a better average reading for the sail as a whole compared to telltales near the top or bottom of the luff.



Know what's coming.

To go fast, you must anticipate changes in the wind and waves that are coming. As Buddy Melges says, you have to 'present your boat for Mother Nature.' In other words, you must look ahead and not just react to things that have already happened.

It really helps to have one crewmember whose main job is to call out impending changes in the wind and waves. If possible, pick someone who will be sitting on the rail with a good view of the course and few other responsibilities. His or her job includes the following:

- Communication of info such as, 'Flat spot for two lengths' or 'Small puff coming in 20 seconds.'

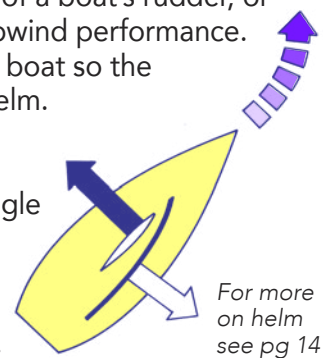
- Talking in a loud, clear voice so the trimmers and helm can easily hear.



Be proactive, not reactive. One of the unique things about sailing is that you must constantly deal with change. The wind changes, the water surface changes, the equipment on your boat changes, and so on. How well you handle these changes will make a big difference in how fast you go. For example, if you always react to changes *after* they happen, you will be going at less than full speed for at least a few seconds every time a change occurs. If, on the other hand, you anticipate changes, you can keep your boat going at or near optimum speed all the time. So keep your head out of the boat and be proactive.

Sail with a bit of windward helm. The feel of a boat's rudder, or helm, is a critical guide for achieving maximum upwind performance. In most conditions, you should trim your sails and boat so the helmsperson feels a slight amount of windward helm.

If he or she lets go of the wheel or tiller, the boat should gradually head up into the wind. Most of the time the 'right' amount of helm is a rudder angle of roughly three to five degrees. This angle helps the rudder create lift without too much drag and it gives the helmsperson a good, positive feel that makes it easier to sail the boat in the groove.



Don't make your 'groove' too narrow. When sailing upwind, it's hard to keep your boat in the groove all the time. As soon as conditions change (e.g. you hit a wave or lull), many sailors lose the groove for at least a little while. That's why it's often good to set the boat up so it has a wide groove, especially in tough conditions like shifty wind and chop. Ways to do this include twisting the sails, easing sheets slightly, making the sails fuller, heeling a little more, footing slightly and so on. This will make the boat forgiving and easier to steer fast a higher % of the time.

Boatspeed

If you're slow, make a change.

A case of the 'slows' is not an insurmountable problem – you just have to try changing things to see what helps. There are many areas you could work on, but start by adjusting a trim variable that will have an immediate and significant impact on performance.

When I'm slow, my first change almost always involves easing or tightening the mainsheet. If this doesn't work, I try the backstay, jib sheet and so on. You'll learn the most if you change only one thing at a time and wait long enough to see its effect, but you don't usually have this luxury while racing.

Trim top batten parallel.

Getting the right mainsail trim is critical for good boatspeed in any condition and in every boat. If you overtrim the main, it will stall and you'll go slow. If you undertrim the main you'll be fast, but you won't point very well.

The key to good performance is finding a nice compromise that works for both speed and pointing. One helpful guideline is to trim the mainsheet so the sail's top batten is parallel with the boom. This works in most conditions, but in light and heavy air you may want to twist the top batten off to leeward a bit. In flat water and moderate air, you can trim the mainsheet quite hard so the sail has very little twist. In these conditions, the top batten may actually be angled to windward.

Top batten
parallel to
the boom



Measure speed relative to other boats.

When you're trying to work on boatspeed, there are a number of ways to gauge how fast you're going. You can use a knotmeter, for example, or simply tune in to the feel of your boat going through the water. But in the end the only thing that matters is how fast you are going relative to your competition – so that's the ultimate measure of performance. For instant and accurate feedback on your speed, grab any chance you can to sail near another boat, and watch carefully how the two boats compare.

Put trim marks on your controls. One of the keys for good boatspeed is the ability to reproduce the way your boat was set up when you were going fast. So make sure all your rig and sail adjustments (e.g. turnbuckles, jib leads, control lines) are calibrated and marked with a numerical scale. This will make it easy to adjust, discuss and record precise trim settings.



Record fast settings. Once you've marked all your trim controls, start keeping a record of how things were set up during races when you were fast (and slow). Your goal is to be able to reproduce fast settings quickly rather than having to reinvent the wheel each time you go racing.

Copy the fast boats. Your competitors are a great source of go-fast ideas, so keep an eye on them. Pay particular attention to boats that are going faster than you, and don't be afraid to copy how are they trimming their sails, for example, and positioning their weight in various conditions. You can learn a lot just by watching them on the race course; many of these sailors will also be happy to share ideas if you talk with them ashore before or after racing.

Pull your outhaul tight. Try sailing upwind with your outhaul a little tighter than normal. It's true that a looser outhaul will add power to the sail, but it also creates a lot of 'return' in the lower leech (i.e. the bottom batten hooks to windward).

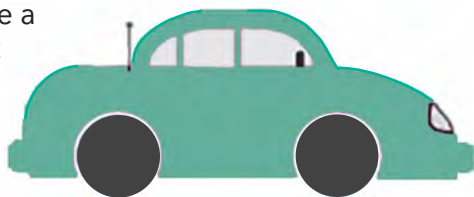
This increases the likelihood of stalling. With a tighter outhaul you can trim the mainsheet harder overall, which is usually good for pointing and speed.

Boatspeed

Change gears – early and often!

When the wind or sea conditions change, as they always do, you must promptly make corresponding changes to your boat and sails by 'changing gears.' This process is critical for maintaining good speed over the course of a race. The fastest sailors are skillful at sensing changes in the wind and waves, and they shift gears a lot (and sooner) compared to less-fast racers.

When you need to change gears, there are many possible adjustments you can make. But many of those are difficult to make or they don't have much effect. It's better to focus on things that are easy and quick to adjust and have a significant impact on performance (like mainsheet, jib sheet and backstay).



Use your sheets to change gears. The best way to change gears quickly, easily and substantially is with your mainsheet (and/or jib sheet). A simple trim or ease affects the attitude of your entire sailplan. So this should usually be your first move.

Don't cleat the sheet! When you are changing gears a lot, try not to cleat your mainsheet. This line gives you a good feel of the boat's pulse, and it's the quickest way to make small changes to keep the boat going fast. When you cleat it, you lose that feel and delay your response to any changes in the wind and waves. Of course, when it's windy the helmsperson may need help holding the sheet, or you may have to put it around a winch. Just try not to use a cleat.

Minimize rudder movement.

Steering a lot with your rudder is generally slow (on any point of sail) because it creates unnecessary drag and turbulence. Even when you're sailing upwind in waves, less steering is usually better. Of course, it can be very helpful to make subtle course adjustments to miss the worst waves. But once you are about to hit the first bad wave, it's usually best to lock your rudder in the center until you're through the worst of it. When you're pounding in waves, the last thing you need is extra drag from a moving rudder.

Are the waves symmetric?

The waves are usually perpendicular to the wind and therefore the same on each tack, but this is not always the case. A wind shift, cross-current or distant storm can create waves that are asymmetrical. The first step is recognizing when this is the case and the second is to trim your sails and boat different as necessary from tack to tack. For example, the wavy tack requires more power and twist in the sails and you can't point as high as on the smoother tack.



Be fast right after the start. A big priority when setting up your boat before the start is to optimize your sails and tuning for the first few minutes after the start. When you come off the line, you want to sail your boat at its absolute fastest so you can poke out in front of the pack. So set up for the wind you have at the start, even if you expect it to change later in the leg. If the wind does change, it's usually better to be leading the fleet with the wrong set-up than back in the pack with perfect sail trim.

Power up before you hit waves.

When you're sailing in waves, it's important to make sure your boat is powered up and going fast when you hit the bad ones. Anticipation is key. The secret to maintaining speed through a bad wave set is shifting gears before you get to those waves. That means you need enough warning to power up your sail plan before the bow digs into the first wave. As soon as you see a bad wave coming, make sure everyone knows. Then ease sheets, bear the boat off for speed, lean your crew weight aft and hold the rudder straight.



More waves than wind?

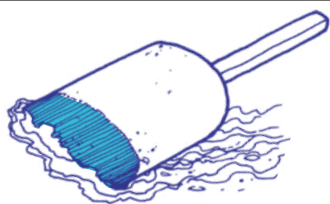
With any wind speed, there is a certain size and shape of waves that you expect to see. But sometimes you get more waves than wind (e.g. a dying breeze), or more wind than waves (e.g. a building breeze). It's important to know what you have so you can make the appropriate trim changes. With more wind than waves, for example, you can trim hard and point high. But when you have more waves than wind, you need speed rather than pointing.

Light Air

Make your boat feel-friendly.

The 'feel' of your boat is a critical factor in making it go fast. When you can feel the forces at work on your hull and sails, it's easier to understand what the boat needs.

Set up your boat so it's 'feel-friendly.' For example, eliminate play in the steering system, and use sheets as light as possible. Then sail your boat more by 'feel.'



Watch out for current.

In light air you don't go very fast through the water, so the speed of the current is a larger percentage of your boatspeed. This means current has a proportionally greater effect on strategic considerations such as wind speed, wind direction, laylines and the favored side of the course. When the wind is light, don't underestimate the effects of current.

Increase windward helm.

Most boats go fastest upwind with a moderate amount (about 3° to 5°) of windward helm. This makes it easier to stay in the groove and helps the boat point higher naturally. When the wind is light, however, it's hard to get enough helm to realize these benefits. If the helmsperson doesn't have much feel, take steps to increase it:

- Move crew weight forward
- Make the boat heel more
- Avoid pinching; sail faster
- Rake the mast farther aft
- Sail with less rig tension
- Make your sails fuller
- Pull the traveler to windward
- Move the centerboard forward
- Trim your mainsheet harder



Prioritize additional pressure.

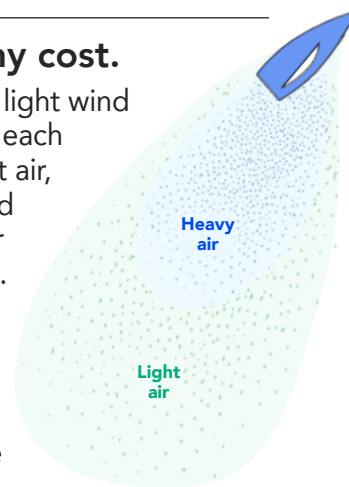
When you're racing in heavy air, a one- or two-knot increase in wind speed might improve your boatspeed slightly, but it probably won't help you point higher. In light air, the same wind increase could improve your speed substantially, and it will allow you to point noticeably higher.

This means puffs are a lot more valuable, relatively, when the wind is light. So, in light-air conditions focus on looking for and sailing in better pressure. This will usually help your performance much more than other strategic factors such as wind shifts or current.

Avoid wind shadows at (almost) any cost.

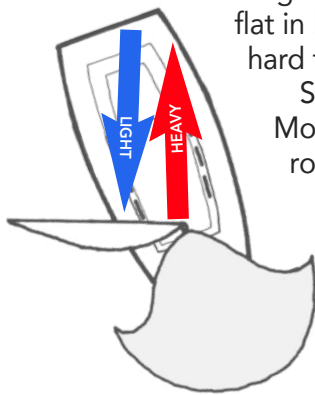
Sailing in bad air is slow, and if you're racing in light wind it's very slow. There are two reasons for this: 1) each boat's wind shadow covers a larger area in light air, so a bigger percentage of the course is covered by bad air; and 2) bad air is more damaging for boats that don't have much wind to begin with. In summary, fight hard for clear air!

Make a light-air checklist. Even some Olympic-level sailors I know use this technique every day they go racing. As a team they make up checklists of important things to remember about speed, strategy, tactics, etc. in various conditions (e.g. light air, heavy air, waves). Then on the way out to the race course (or between races if things change) they review the checklist for that day's conditions.



Move your weight for pressure changes.

Wind pressure can have a huge influence on the ideal location of crew weight during beats and runs. In most cases, of course, you should move to windward when it's windy and to leeward when it's light. This is largely a matter of controlling heel angle and therefore helm. If the boat is too flat in light air, you won't have much helm and it will be hard to stay in the groove.



Similar reasoning applies to fore-and-aft weight. Moving crew weight forward puts more of the boat's rounded bow section in the water and raises some of the flatter stern out of the water. This is good in light air because it increases windward helm and reduces wetted surface. Moving crew around for different wind speeds is obviously very important for small, lightweight boats, but it will help you 'change gears' on any size boat.

Downwind

Plan ahead for the run.

You should have a strategic plan in place when you begin the run, so start thinking and talking about this well before rounding the windward mark. Do you prefer left or right on the run? Will you do a jibe set or a bear-away? Make sure everyone on your team is aware of the plan so they know what maneuvers will be needed around the mark.



Steer for low spots. When you're sailing downwind in waves, you can gain a lot of distance by surfing better than your competitors. You have to adjust your sails, weight and course continually to surf down the fronts of the waves and stay on each wave as long as possible. One of the best surfing tips I ever got was long ago from a top sailor who told me to 'steer for the low spots.' The basic technique is to look at the wave that is at, or just ahead of, your bow and find where that wave is lowest. Aim for this spot and you will always (or at least usually) be sailing downhill.

Sail away from the next shift. When you're racing upwind you sail *toward* the next windshift you expect because when it comes you'll be on a higher ladder rung with less distance to sail to the windward mark. But on a run you should sail *away from* the next shift because you're trying to climb *down* the ladder rungs to the leeward mark.

By sailing away from the shift, you position yourself so when the shift comes you can jibe and sail on the headed tack (which is fast). One exception to this rule of thumb is when the windshift also brings an increase in pressure. Because added pressure helps you sail faster and lower, it often pays to sail toward this new wind, not away from it.

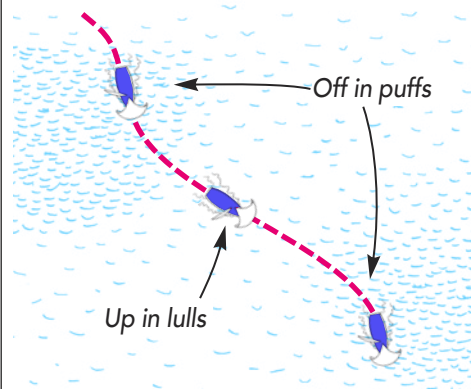
Jibe on the lifts. When you're racing upwind in an oscillating breeze, you tack on the headers to sail on the lifts. But on runs the best strategy is to jibe on the lifts so you are always sailing on a header. By staying on the headed jibe, you sail lower and aim closer to the leeward mark.

Jibing on lifts follows the downwind strategy of sailing away from the next shift. If you're on port tack and you get lifted, that means the wind shifted left. The next shift will come from the right, so you should jibe and head left. Then you'll be on a lower ladder rung when the shift comes.

Of course, you have to be careful about playing too many shifts if jibing is costly, such as in heavy air, light air or with a new crew. As with any racing move, the benefit of each jibe has to outweigh its cost.

Go up in lulls, off in puffs.

This may be the most valuable piece of advice for any offwind leg, and it's easy to use. By heading higher in the lulls, you keep your speed high so you can get through the lull quickly to the next puff. By bearing off in puffs, you maximize VMG to leeward and you sail more in the direction the puff is traveling, so you realize the benefits of the puff for a longer time.



Minimize helm downwind.

When you sail upwind, it's good to have a bit of windward helm to give the rudder more feel and lift. But when sailing downwind you don't need this feel, and an off-center rudder just creates drag, which is slow. Therefore, adjust the trim of your boat and sails to keep your rudder centered and your helm neutral as much as possible. The boat should basically steer itself.



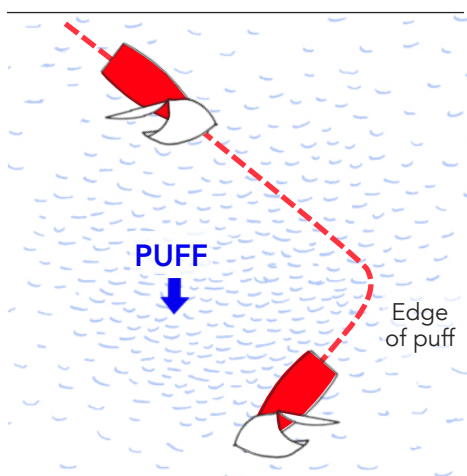
Avoid 'clumps.'

Other boats give you bad air and waves and block you from going where you want. These effects are multiplied in a pack, so stay away from clumps.

Downwind

Bear off to stay in puffs.

More wind is almost always better, so try to stay in puffs as long as possible. A good way to do this is by sailing lower so you move in the same direction as the puff. The puff is usually going straight downwind, so try to match this angle as much as possible. Even if dead downwind is lower than your optimal velocity-made-good (VMG) angle for that wind speed, it will be a gain overall if it keeps you in the puff longer (compared to a higher, faster angle that takes you out of the puff sooner). This technique is especially good for boats with symmetrical spinnakers (or no spinnakers) because they normally sail low angles and can often sail lower without losing too much VMG.



Jibe to stay in puffs.

Bearing off with a puff is not always the best option. When the wind is light or your boat doesn't like to sail deep downwind, it's usually best to keep sailing a higher angle. But when you get to the edge of a puff (where you feel less pressure and there is more wind astern than ahead), don't keep going.

Instead, jibe so you sail back through the puff and keep the extra wind speed. This works well for fast boats and boats with asymmetrical chutes that usually sail high angles and go slowly if they sail too low.



Trim your chute with a curl.

When you're flying a spinnaker (either asymmetric or symmetric), trimming the sheet too much is slow. To make sure you have the sail eased out far enough, keep easing your sheet until you see a slight curl along the luff of the spinnaker, and carry this curl for most of the reach or run.

Though this technique does give up a slight bit of sail area, it's much faster to have the sail eased all the way with a small curl than to get rid of the curl but risk having the sail overtrimmed some of the time.

Use your wind shadow to attack.

If there is ever a 'silver lining' to being behind, it's on a run where you can use the long reach of your wind shadow to slow or control boats ahead. There are basically two ways to do this:

- Position your wind shadow directly on the other boat (using your apparent wind angle as a guide) to slow them down. This works best in lighter air where your bad air extends farther and has more of a slowing effect. Of course, the other boat can always head up or jibe to escape.
- If it's difficult to capture (or keep) the other boat in your wind shadow, you may be able to use your bad air to affect where they are able to sail. For example, you could place your wind shadow between the boat ahead and the favored side of the run. This would give them a tough choice: sail to the wrong side in clear air, or to the favored side in bad air.



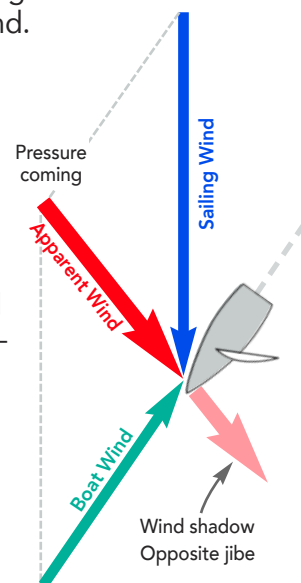
Watch your apparent wind.

Your apparent wind is the wind (direction and speed) that you feel on your face and that is shown by your masthead wind pennant and the telltales on your shrouds. It happens to provide a lot of good information and guidance when you are sailing downwind.

Pressure that's coming. When you're trying to identify the puffs (and lulls) that will come to you in the near future (assuming you stay on the same jibe), simply look in the direction of your apparent wind. What you see is what you will get.

Location of wind shadows. A big challenge on runs is keeping your air clear. Locate other boats' wind shadows by looking at their masthead wind pennants – their bad air goes to leeward in that direction. Also, your own telltales point to your wind shadow, which is helpful to know when trying to slow a boat ahead.

Opposite jibe angle. On many boats, the inverse of your apparent wind direction is a good indicator of what your heading will be on the opposite jibe.



Boathandling

Minimize maneuvering.

When you have a good reason to tack or jibe, go for it. But if you don't, there are a couple of reasons why you should err on the side of minimal boathandling. First of all, every maneuver you make costs distance. A tack, for example, may cost as much as two or three boat-lengths, especially for heavier boats in light air and chop. If you do even a couple extra tacks on a beat, you can lose significant distance.

Another reason to minimize the number of tacks and jibes is that they involve a certain degree of risk. When you tack, for example, you might capsize or get an override on your genoa winch. A jibe could result in a torn or tangled spinnaker. By avoiding unnecessary maneuvers, you will generally be faster and safer.

Develop a team playbook.

One of the toughest things for any racing sailor is hopping on a new boat and trying to 'learn the ropes' while in the heat of battle. You can make this a lot easier for new members of your team by creating a team playbook that describes each position on the boat, including their specific responsibilities during all possible maneuvers. Ask all your current team to help compile this for their position. Put it online (so it's easy to access, update and send to new people) and keep updating!



Tacking



Don't eat your sheet. When you are trimming your mainsail, do NOT hold it with your teeth. This is a painfully slow and ineffective way to trim the sail, and of course, it's terrible for your teeth. Instead of your teeth, use the hand that is holding the end of your hiking stick. This allows you to pull the sheet in hand over hand (not hand over teeth)!



Customize your tacks.

In static conditions, try to make every tack identical. But when things change (boat type, wave size, wind speed), you must also change your tacking technique.

Type of boat – Heavy boats have more momentum, can carry their speed through a tack, and are hurt by sharp turns, so their tacks are generally slower. Small, light boats don't glide much so they turn more quickly; they also accelerate faster so quick turns don't hurt as much.

Wave size – A good rule of thumb for tacking in waves: The bigger the waves, the more quickly you should turn the boat. Hitting waves head on is slow, so don't stay head to wind (pounding straight into the waves) in the middle of a tack any longer than necessary. In flat water, it's OK to make a more gradual turn through the wind.

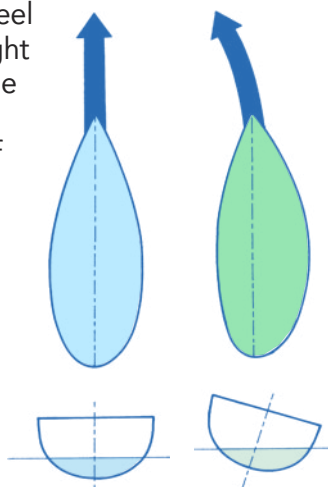
Wind speed – Wind, like waves, slows you a lot when you're heading straight into it. So don't linger in the middle of a windy tack. If turning quickly makes it tough for the crew to cross the boat and get the sails in, stop your turn slightly above closehauled to give everyone a chance to get settled.

Use crew weight to steer, especially in light air.

Whenever you turn your rudder away from centerline, it creates drag and slows you down. So, before turning your tiller or wheel, move your crew weight to change the boat's angle of heel and help steering. This is especially effective in light air and smaller boats, but it will help at least a little in almost any boat or condition.

When your boat is flat (not heeled), the part of the hull in the water is symmetrical (*left drawing*). This makes the boat continue straight ahead. But if you heel the boat to starboard, the underwater shape is no longer symmetrical (*right drawing*), and the boat will turn to the left.

If you want to turn to port, heel the boat to starboard. If you want to turn to starboard, heel the boat to port. This allows you to turn the boat with less rudder angle, which is better for speed.



Boathandling

Build speed for maneuvers.

Whenever you turn a boat to make a maneuver, you lose precious boat-speed. If you start the turn at less than full speed, your speed loss will be compounded and it'll be tough to accelerate afterward. That's why you should try to be at full speed (or even a little faster) before you begin any maneuver (e.g. a tack).

Don't maneuver in lulls.

Because it helps to have speed when you maneuver, try not to make turns when there is less wind pressure. Otherwise the cost of the maneuver will be multiplied. When possible, time your tacks and jibes so you perform them in good wind pressure. Of course, if you're racing in lots of breeze it may be safer (i.e. less risky) to maneuver in lulls rather than puffs. In that case it's sensible to wait for less wind to make turns.



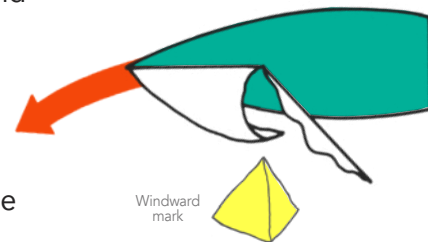
Jibe when you're going fast.

When the wind is blowing hard enough to make jibing risky, it seems that the safest maneuver may be to jibe when you're going relatively slowly. But the opposite is actually true – it's better to jibe when you are sailing as fast as possible. Speed has two advantages: 1) it makes your hull a more stable platform; and 2) it reduces wind pressure on your sails (when sailing with the wind, more speed equals less apparent wind). So when jibing in heavy air, think of speed as your friend. One of the best times to jibe is while surfing down a wave.

Use sail trim to steer, especially in heavy air.

In heavy wind, moving your weight around doesn't have much effect on the boat's angle of heel, especially in bigger, heavier boats. So that's not a useful way of steering. Instead, focus on sail trim. If you've ever tried to bear off around a breezy windward mark without easing the mainsheet, you know what I mean.

When you want to head up, trim in your mainsail so you put more wind pressure on the aft part of your sail plan. This pushes the stern farther away from the wind and, since the boat rotates on its keel or center-board, makes the bow head up. Conversely, when you need to bear off (e.g. at the windward mark), ease your mainsheet to allow your bow to turn away from the wind. The same principles apply to jib trim – pull it in to bear off and ease it when heading up (e.g. at the leeward mark).



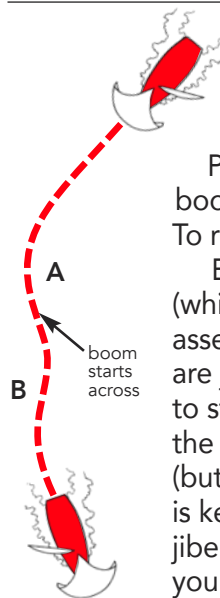
Ease your main to help the bow turn down.

Make an S-turn jibe in heavy air.

The scariest (and riskiest) thing about jibing in heavy air is usually the possibility of capsizing or broaching.

Problems can occur while you're struggling to get the boom across, or right after your sail fills on the new jibe. To reduce your chances of disaster, make an S-turn jibe.

Begin an S-turn jibe just like most jibes by bearing off (while going down a wave, if possible). Make your turn assertively (A), and don't hesitate in the middle. When you are just about dead downwind, pull very hard on the boom to start it coming across. As soon as you are 100% sure that the boom is going across, turn quickly back the other way (but not so far that you go by the lee on the new jibe). This is key because it unloads the mainsail as it fills on the new jibe. When the boat is under control, head up gradually to your optimum angle on the new jibe.



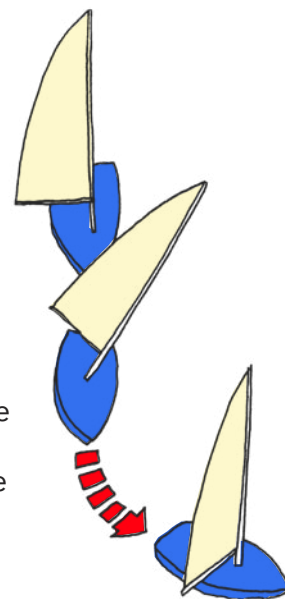
Steer under your mast.

When you're sailing downwind in breezy and/or wavy conditions, the best way to stay in control (and keep going fast) is to steer your boat under the rig and sails. If the top of your rig goes to windward, head up to get the hull under the mast.

If the top of your rig goes to leeward, bear off. The idea is to keep the mast roughly vertical by steering your boat so the bottom of the mast stays under the top. It's like trying to balance a long stick vertically with one end in the palm of your hand.

If you have a symmetrical spinnaker, an easy guide

is to aim your bow at the center of the sail's foot (where there is often a vertical seam).



At Marks

Find the next mark.

One easy way to lose a race is by sailing to the wrong mark, or by sailing in the wrong direction when you can't find the next mark. Be sure to read the sailing instructions so you understand the course and what the marks look like. Then, before every race, figure out the compass course to each mark.

During the race, locate each mark visually before you round the previous mark. I usually assign this job to one crewmember. As we approach the windward mark, for example, his or her job is to find the leeward mark (and show the rest of us). This is essential for implementing your strategic and tactical plan.



Drop your chute early.

One of the most common and costliest mistakes at leeward marks is leaving your chute up too long. Dropping your spinnaker a few boatlengths early will have almost no effect on the speed of most boats. Therefore, unless you really need to fight for an inside overlap, make the smart, conservative choice to douse a little early and avoid risking a race-ending foul-up.

Remember risk.

Marks usually involve a relatively high degree of risk because all the boats converge there, the fleet gets very compressed, and the stakes are high. So before you get to a mark it's helpful to think a little about how much risk are you willing to take. For example, would you be willing to try cutting inside the boats ahead if that meant a 50% chance of fouling? What if it was only 20%?

How much risk you take at a mark depends on a few factors including: a) how far you are from the finish; b) how you're doing in the race; and c) how you're doing in the series. If you're at the first mark of the first race in a series, for example, you shouldn't take big chances. But if it's the last mark of the last race, that could be a different story.



Make a 'tactical' rounding in traffic.

The classic way to round a jibe or leeward mark is to make a wide approach on the near side so you can cut close on the far side. By swinging wide at first, you create space for a smooth and fast luff up to a closehauled course. By cutting close on the far side of the mark, you gain maximum distance to windward as you exit the mark. This is key for avoiding the wind shadow of the boat in front of you.



Make a 'strategic' rounding by yourself.

When there's not a boat close in front of you, gaining maximum height when exiting the mark is not a priority. This means you don't have to swing so wide on your approach; in fact, the fastest course around the mark is one where you are on a beam reach as you go by the mark. In this strategic rounding you sail only as far to leeward as necessary.

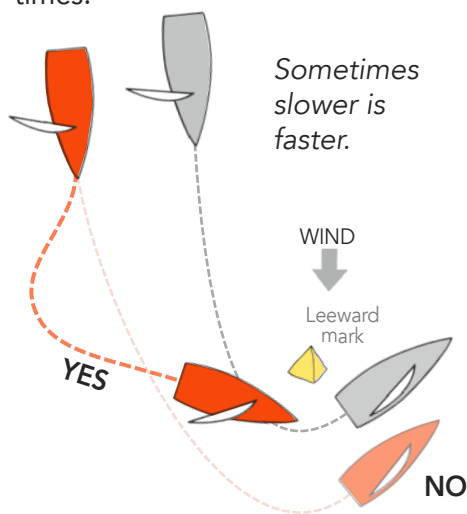


Play to the inside before the mark. When you're sailing down the run, you often have a choice between sailing on port jibe toward the left side of the course (looking downwind) or on starboard jibe toward the right side. At the beginning of the leg, make this choice primarily on strategic factors like wind pressure and shifts.

As you get closer to the leeward mark, however, think more tactically. At this point, strategic factors don't matter as much (because the boats aren't as far apart), so you must go for position relative to other boats. When rounding the mark to port, it's strong to play the left side of the course (looking downwind) as you get near the mark. This will give you inside position and the right of way as you converge with other boats.

At Marks

Slow down to go faster. In most racing situations, faster is better because the goal of racing is to get to the finish line as quickly as possible and beat the other boats. But once in a while you need to slow down. Rounding the leeward mark is a perfect example. If you round on the outside of another boat, you will often be much worse off than if you slowed down and rounded right behind them. Don't be afraid to go a little slower at times.

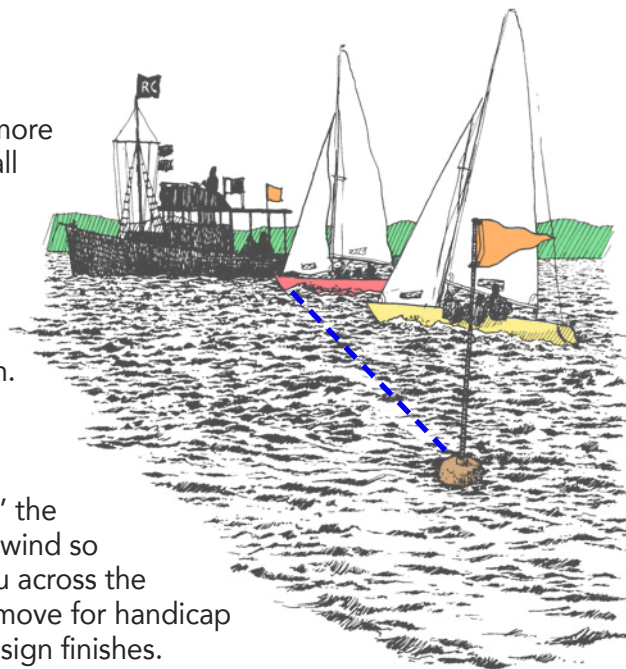


Race to the zone. On a run, the leeward mark is your ultimate goal, but you should actually be racing to the windward edge of the zone. Once boats enter the zone, their rounding positions are essentially established by rule 18 (Mark-Room). So aim to have maximum advantage when you get to a point that's about three lengths to windward of the leeward mark. In most cases, the boat that gets to the zone first will round the mark first.

Jibe before the mark. Often you have to jibe to round the reach mark or leeward mark. But jibing while you are turning around a mark can be a tough maneuver. To make this a lot easier, sail wider as you approach the mark and do your jibe before you get there.

'Shoot' the finish.

You can often catch a few more boats if you keep pushing all the way to the finish line. One end of the line is almost always 'favored' (e.g. farther downwind at an upwind finish), so that's where you should aim. Make your final approach about a boatlength or so below the layline to this favored end, and then 'shoot' the finish by luffing up head to wind so your momentum carries you across the line. This is always a smart move for handicap racers and for close one-design finishes.



Delay non-essential clean-up.

The most crucial part of any mark rounding may be the minute or so just after you round the mark. That is when you're fighting to get the boat going fast in the right direction amidst a bunch of bad air and choppy waves. During this time it's critical to focus 100% on sail trim, weight placement, steering, tactics and strategy. This is not a time when you want half the crew (or even one of them) to be distracted with boat clean-up.

Once you've rounded the mark, allow one person to perform only jobs that are absolutely essential – then focus entirely on the race.

Leave a gap in front at the leeward mark.

When you're rounding the leeward mark behind another boat, being closer is not always better. Having your bow close to the other boat's transom may feel good, but it creates two possible problems: 1) If the other boat slows unexpectedly during the rounding, you may have nowhere to go but to leeward and outside of them, which is very slow; and 2) if you exit the mark right behind the other boat it will be very tough to escape their bad air. For these reasons, it's usually better to round a full boatlength, rather than a few feet, behind the boat ahead.



Beware the port-tack layline.

Approaching a windward mark on port tack is often a good way to sneak in front of boats that are jammed up in bad air on the starboard layline. But it's risky, especially if you're in the middle of a pack, because you are a give-way boat on port tack. One way to reduce risk is by making your port-tack approach farther from the port layline (at least 4 to 5 lengths below it) so your tack on the starboard layline is outside the zone.

Tactics

Minimize tactics.

We spend a lot of time talking about good tactical moves we can make on the race course, but that doesn't mean we should actually use them. Every time you make a maneuver relative to one or two other boats, you lose distance and time to every other boat in the fleet. So develop your tactical arsenal, but use it only when essential to follow your strategy or beat another boat.



'Spend' your lead.

In one-design racing, it doesn't matter how far you are ahead of other boats when you cross the finish line, so don't take chances trying to maintain or build a large margin. In fact, it's often smart to 'spend' (i.e. use up) some of your lead in order to consolidate your position and make sure you beat the boats behind. Of course, handicap racers would usually not do this.

Know the score. When you are trying to make a strategic game plan and assess the amount of risk you're willing to take, you have to know where you stand in the fleet. If it's the last race of the regatta and you're a close second overall, for example, you will likely have a very different approach than if you are 10th. When the score might make a difference in how you sail a race (usually late in a series), be sure to write down the overall scores and bring them on the water with you.

No surprises!

Surprises are fun at birthday parties, but when you're racing sailboats your guiding philosophy should be 'No surprises.' When something happens by surprise it means you were not expecting or prepared for it, and likely you didn't have much time for deciding how to respond. We all know that decisions are better when you have more time to make them, so look and think ahead. Some things (e.g. wind shifts) will always surprise us occasionally, but the actions of other boats (and your resulting tactical moves) should seldom be surprising.



Make 'contingency' plans.

In sailboat racing, you don't have control over the actions of other boats. For example, a boat that's to leeward and ahead on port tack might suddenly tack and have the right of way on starboard tack. If you aren't prepared for this, you can easily break a rule or end up sailing the wrong way strategically.

One way to handle unpredictable events like this is with 'contingency planning.' Anticipate situations that may occur in the near future, and ask yourself (and your crew), 'What will we do if _____?'

In the example above, it's much better to plan your response while that boat is still on port tack rather than waiting until she tacks and you have one second to decide.

Read World Sailing Case 50.

The right-of-way rules govern your interactions with other boats and therefore fall under the category of 'tactics.' One of the most commonly used (and broken) rules is rule 10 (Port - Starboard). If you want some good guidance about this rule you should check out World Sailing Case 50, which says in part:

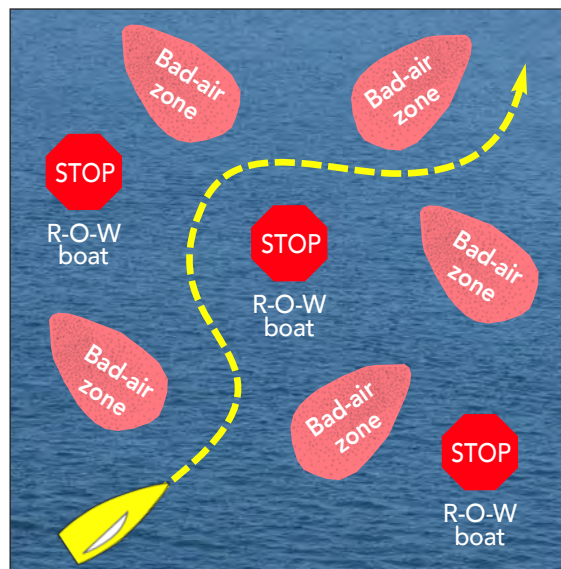
'... When the committee finds that S [the starboard tacker] did change course and that there was reasonable doubt that P [the port tacker] could have crossed ahead of S if S had not changed course, then P should be disqualified.' This case should be required reading (and a warning) for anyone who is on port tack thinking about crossing close in front of a starboard tacker.

Avoid other boats!

There is almost never anything good that comes from being near other boats in a race. Potential problems with boat traffic include:

- Keeping clear of boats with the right of way;
- Being slowed by sailing in (or trying to avoid) wind shadows and bumpy water;
- Losing your option to tack or jibe when necessary.

For all these reasons, try generally to avoid getting tangled up with other boats.



Tactics

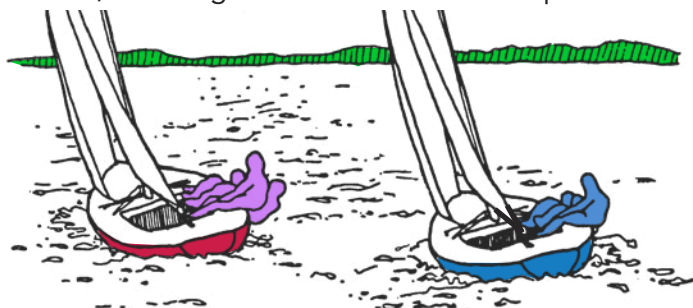
Choose tactics based on strategy.

Strategy is usually your top priority because it concerns the big picture and often has a huge impact on your race results. Tactics are the boat-on-boat moves you make to implement your strategy. The normal procedure is to make a strategic plan and then use tactics (if necessary) to pursue it. This port tacker, for example, should first decide if she wants to go left or right; then she picks a tactic (duck or lee bow) to go there.



Watch out for wind shadows. When you're trying to go fast upwind, bad air is your enemy. So watch out for other boats (look at their telltales to know where their wind shadows are going), and plan ahead to get lanes of clear air. Wind shadows extend farther and slow you more in light air than they do in heavy air. So when there's not much wind, don't stay in a position that's to leeward of or behind another boat. In breeze, however, it might be OK to sail in a boat's bad air, especially if you're on a lift or going the right way.

Cover the boats behind? 'Covering' is a tactic that helps you stay ahead of boats behind. The basic idea is to draw an imaginary line between the boat(s) you want to beat and the next mark. Position your boat on that line so you are between your competition and the mark – this will make it tough for them to get past you. However, you don't always want to cover. You should ignore boats behind when: a) it's early in the race; b) you're trying to pass boats ahead of you; or c) you're on a lift, or sailing toward the next shift or puff.



Use a blocker to hold your lane.

When you want to keep sailing upwind on starboard tack, use a 'blocker' to protect your lane. A blocker is a boat on starboard tack that's to leeward and ahead of you, in a position to intercept incoming port tackers. Those boats either have to lee-bow the blocker or duck behind the blocker (and go behind you too), leaving you free to sail with clear air in your lane.

Be willing to 'bite the bullet.' It's never easy to cross behind other boats on a beat since this is a clear concession that they are ahead of you. But when you have a persistent windshift, sometimes you have to 'bite the bullet' (i.e. face the unpleasant reality of passing behind them). As soon as you realize what the wind is doing, get on the tack that takes you toward the persistent shift and keep going, even if it means crossing behind your competitors. This may seem painful at first, but you will gain in the long run.



Wave a port tacker across.

When you have a good lane of clear air on starboard tack and you want to keep going toward the left side, don't automatically yell 'Starboard' to every port tacker that tries to cross in front of you. If a boat tacks on your lee bow it will be difficult to keep going left, so consider letting the other boat cross in front of you, even if this means you will lose a little by bearing off behind them. When you are still several lengths away, yell 'Go ahead, Cross' (or words to that effect) and wave your arm to let them know they can cross.

Pass one boat at a time. When you're playing catch-up from behind, don't try to make up your deficit all at once. This is risky and usually not possible. It's better to be conservative and methodical, picking up one or two boats at a time, rather than aiming for one large gain with a small chance of success. Go for a lot of smaller gains, each with a better probability of working. Be like the tortoise instead of the hare, especially early in a race or series.

Keep your options open. Whether you're racing upwind or downwind, you have only two choices at any moment: sail on starboard tack or sail on port tack. It's important to keep both options available during as much of a race as possible since you never know when you might have a strong reason to tack or jibe. So don't let another boat get into a position where they are blocking your ability to follow your strategy.

Learning

Minimize mistakes.

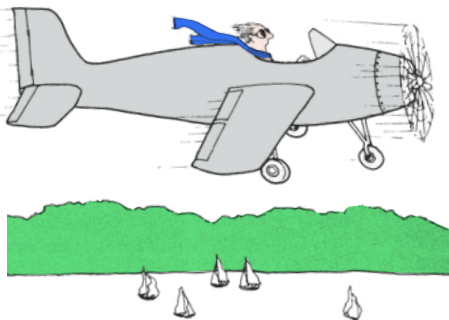
It's very difficult, if not impossible, to sail a perfect race, but you should always try to minimize how many mistakes you make. This is especially true when you're behind and trying to catch up. Be patient, don't take unnecessary risks and try to consistently make good decisions. Let other boats make the errors and you'll have a great chance to pass them.

'Start' the race again. If you often sail the first part of a race pretty well but seem to lose boats after that, here's an idea: When you get to the point where things typically go south, pretend the race is starting all over again. In this new race, your main goal is to 1) avoid losing any boats before the finish; or 2) see how many boats you can pass before the finish. Keep track of how well you do, and see if this helps your overall results.

Remember the big picture. A large percentage of errors made by racing boats occur because sailors are too focused on what's happening in or near their own boat. A good example is failing to notice that you could fetch the next mark on one tack. Or not seeing a converging starboard tacker until the very last second. Remind yourself (and your teammates) to keep your head(s) out of the boat.

Drop in to the rescue.

One of the toughest things about coming from behind

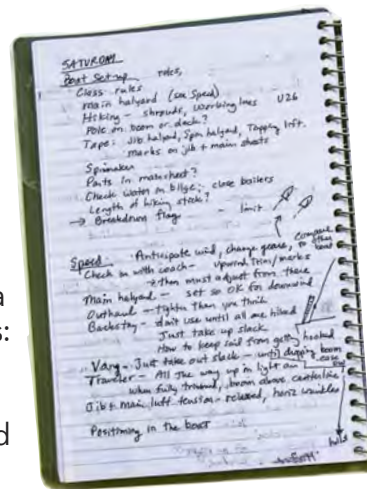


is psychological. The key is to stay focused on the present and not get hung up on how you ended up near the back of the fleet. Here's a psychological game I play when I find myself back

in the pack: I pretend that *someone else was driving the boat and got it into a bad position*. Then I magically dropped onto the boat from an airplane to help make a comeback. This does a number of things: 1) it keeps me from blaming myself for making a mistake; 2) it helps me look at the current situation objectively; and 3) it gets me psyched for the challenge of improving my position during the rest of the race.

Keep a notebook.

Keep a written record of things you learn. This will reinforce your learning and make the info available for easy future reference. Use a waterproof book (or tablet) on the boat and transfer your ideas later to a master file with these topics: Boat improvements, boat-speed ideas, tactical notes that worked (or didn't), wind and weather trends, crew roles for boathandling, etc.



Learn something from every race.

You can't win every race, but you can always learn something while you're racing. So keep trying to win, but develop a learning attitude for yourself and your fellow crewmembers. Spend time practicing, keep a daily journal, talk with the top sailors, learn from your mistakes, and most importantly, recognize things you do well. Try to see at least something you're doing a little better each day you go racing. If you focus on the process of sailing better, your results will inevitably improve, and you'll enjoy the journey too.



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